

PC-02 · SILICONES · PERSONAL CARE

HS 200-12500

Dimethicone

INCI Dimethicone	CAS 63148-62-9	EC —	FAMILY PC-02	SUPPLIED AS Fluid, 100% silicone
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01 POSITIONING

One chemistry, a viscosity ladder — specified by grade, delivered identical, lot after lot.

02 HOW IT WORKS

- 01 Viscosity = function
350 cSt spreads and de-tackifies; 12 500 cSt films and conditions. The grade is the formulation decision.
- 02 Surface leveling
Fills cuticle and skin micro-relief, converting roughness into measurable slip.
- 03 Inert backbone
Si–O chain: no oxidation, no rancidity, compatible with virtually every oil phase.

03 APPLICATIONS

- Rinse-off and leave-in conditioners
- Serums and anti-frizz
- Skin creams (tack reduction)
- Color cosmetics
- Industrial defoaming and release

04 FORMULATION GUIDE

USE LEVEL	0.5–5%
PH & CONDITIONS	pH-independent; shear-stable
— Add to the oil phase or post-add under moderate shear; pre-blend high-viscosity grades with a light ester for easier incorporation.	
— For rinse-off deposition, pair with cationic conditioning polymers.	

05 TYPICAL SPECIFICATION

Appearance	Colorless clear fluid
Viscosity (25 °C)	per grade: 350 / 12 500 cSt
Volatiles (150 °C)	≤ 0.5%
Purity (Si backbone)	cosmetic grade

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Combing force reduction, wet (class data)
With dimethicone system

Gloss uplift
+ 1–2% dimethicone

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.